

Research Statement

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1 Introduction

My research has mainly focused on the modeling of liquid crystals (LCs) using the cholesteric Landau-de Gennes (LdG) model, and numerical analysis of this model. In particular, in my doctoral dissertation [1] and published paper [2], I focus on a sub-type of LC which are known as *cholesteric nematic LCs*, where the molecules have a chiral structure that affects their actuation and equilibrium behavior.

In my thesis, I presented several new results in the numerical analysis of this cholesteric model. Namely, I proved certain restrictions on the time-step and mesh-size that are not widely found elsewhere in the literature. I have also created a code which computes this numerical analysis and solves the LdG, with these constraints in mind. Simulation results are presented later in this statement.

At my current position at Carnegie Mellon University, I have recently started a research project involving porous phase fluid flow, and plan to implement a version of Uzawa's algorithm to solve it.

2 Landau-de Gennes Model of Liquid Crystals

2.1 Liquid Crystal Theory

Liquid crystals consist of rod-like molecules which allow them to assume a state that possesses the qualities of both solids and liquids simultaneously. If we zoom into any given point in an LC, we see that the molecules there thus have a general direction in which they are pointing, as well as an orderliness to this alignment. And so the LdG theory [3] introduces a *tensor-valued* function Q to describe the local order (at each point in space) in the LC material. Most commonly, we have the so-called *uniaxial* state, where Q may be expressed as

$$Q = s \left(\mathbf{n} \otimes \mathbf{n} - \frac{1}{3} I \right), \quad (1)$$

where s is called the *degree-of-orientation*, and is a measure of the orientational order of the LC molecules at each point, and $\mathbf{n}$ is called the *director*, which has unit length, $|\mathbf{n}| = 1$, and represents the average direction in which the molecules are pointing.

Let Ω be the physical domain of interest, assumed to have Lipschitz boundary Γ with outward pointing unit normal vector ν . The tensor Q belongs to the space $\mathbf{S}_0 = \{Q \mid \text{tr}(Q) = 0, Q^\dagger = Q\}$. The free energy of the LdG model is defined as [4]:

$$\begin{aligned} \mathcal{E}[Q] := & \int_{\Omega} f(Q, \nabla Q) dx + \frac{1}{\eta^2} \int_{\Omega} \psi(Q) dx \\ & + \frac{1}{\omega} \int_{\Gamma} g(Q) dS(\mathbf{x}) + \int_{\Gamma} \phi(Q) dS(\mathbf{x}) - \int_{\Omega} \chi(Q) dx, \end{aligned} \quad (2)$$

where f is the elastic function (quadratic) dependent on the material parameters ℓ_1, ℓ_2, ℓ_3 , and the twist coefficient τ_0 , which is very important to the cholesteric model, causing stripe-like patterns to emerge, which have a period of $p = 2\pi/\tau_0$ (see Figure 1). The bulk potential ψ is a quartic, double-well type of function, dependent on material parameters a_2, a_3, a_4 . The order parameter that minimizes ψ is s_0 . The surface energy, composed of the quadratic $g(Q)$ and higher-order $\phi(Q)$, accounts for *weak anchoring* of the LC (i.e. penalization of boundary conditions). This weak anchoring takes the form of *homeotropic anchoring* (where Q is penalized to have directors facing normal to the surface of Ω) and *planar degenerate anchoring* (where Q is penalized to face tangent to the surface). The function $\chi(\cdot)$ accounts for interactions with external fields. Finally, η and ω are constants resulting from the non-dimensionalization of the energy. They become important in the numerical analysis, as they control the nonlinear terms.

2.2 Discrete Gradient Flow

We use a numerical scheme for minimizing (2) by first discretizing in time by minimizing movements [5]. Let $Q_k(x) \approx Q(x, k\delta t)$, where $\delta t > 0$ is a finite time-step, and k is the time index. Then we can minimize the energy using a sequence of elliptic problems: Given Q_k , find $Q_{k+1} \in \mathbf{V} := H^1(\Omega; \mathbf{S}_0)$ such that

$$\delta t^{-1} (Q_{k+1} - Q_k, P)_{\delta t} = -\delta_Q \mathcal{E}[Q_{k+1}; P], \quad \forall P \in \mathbf{V}, \quad (3)$$

where we have defined a time-stepping inner product $(\cdot, \cdot)_{\delta t}$ by

$$(P, T)_{\delta t} := \left(\frac{1}{\eta^2} \int_{\Omega} P : T + \frac{1}{\omega} \int_{\Gamma} P : T \right). \quad (4)$$

Starting from an initial guess Q_0 , we iterate (3) until we reach a final iteration index or some other stopping criteria. In my thesis, I prove the following theorem, which gives a time-step restriction to ensure energy decrease.

Theorem 1. *The sequence $\{Q_k\}_{k=0}^{\infty}$ defined by the method above is monotonically energy decreasing, i.e.*

$$\mathcal{E}[Q_{k+1}] \leq \mathcal{E}[Q_k],$$

provided that

$$\delta t \leq 2 / \max\{a_2 + a_3^2/a_4, 2s_0^2/3\}. \quad (5)$$

It should be noted that this is not dependent on τ_0 .

2.3 Finite Element Method

We approximate (3) by a finite element method; in particular, we approximate Q by a finite element function $Q_h \in \mathbf{V}_h$, where $\mathbf{V}_h$ is a finite element discretized subspace of $\mathbf{V}$. Using the same time-stepping procedure from before, applying it to Q_h : given $Q_{h,k}$, we find $Q_{h,k+1} \in \mathbf{V}_h$ such that

$$\delta t^{-1} (Q_{h,k+1} - Q_{h,k}, P_h)_{\delta t} = -\delta_Q \mathcal{E}[Q_{h,k+1}; P_h], \quad \forall P_h \in \mathbf{V}_h, \quad (6)$$

We iterate this procedure until some stopping criterion is achieved. Since $\mathbf{V}_h \subset \mathbf{V}$, the same arguments in Section 2.2 still hold when replacing $\mathbf{V}$ by $\mathbf{V}_h$. Therefore, the same time-stepping restrictions apply to the fully discrete formulation.

We seek to derive an error estimate for a local minimizer of the LdG energy. The main theoretical result for this, and indeed my most important contribution to the LC literature, is the following.

Theorem 2. *Let $Q \in \mathbf{V}$ be a local minimizer of $\mathcal{E}[\cdot]$, and let $Q_h \in \mathbf{V}_h$ be a local minimizer of $\mathcal{E}[\cdot]$. Then given the proper assumptions (such as continuity, coercivity, regularity, etc.), there exists $h_0 > 0$ and c^* such that for all $h \leq h_0$*

$$\|Q - Q_h\|_{1,\Omega} \leq c^* \inf_{P_h \in \mathbf{V}_h} \|Q - P_h\|_{1,\Omega}$$

The explicit form of h_0 is

$$h_0 = \left(\frac{\alpha_1}{k + (a'/\eta^2) + (s''/\omega^2)} \right)^{1/2} \frac{1}{\sqrt{8}(c_0 + \beta'_1/\eta^2 + \beta'_2/\omega)c_2c_3}, \quad (7)$$

where $k = \max((216\ell_1^2\tau_0^2/\alpha_1) - (\alpha_1/2), 0)$, $c_1, c_2, c_3, \beta'_1, \beta'_2, a',$ and s'' are constants depending on the domain, and

$$c_0 = \ell_1 + 3\ell_2 + \ell_3 + 4\ell_1\sqrt{27}\tau_0 + \beta_3(w_0 + 3w_1) \quad (8)$$

is the constant such that $|a(Q, P)| \leq c_0\|Q\|_{1,\Omega}\|P\|_{1,\Omega}$, with β_3 also depending on the domain.

This theorem is very useful; namely, it shows that higher values of τ_0 (corresponding to smaller periods of stripes in the LC structure) need a smaller mesh size to guarantee a solution sufficiently close to the true solution. This is also true of ω and η .

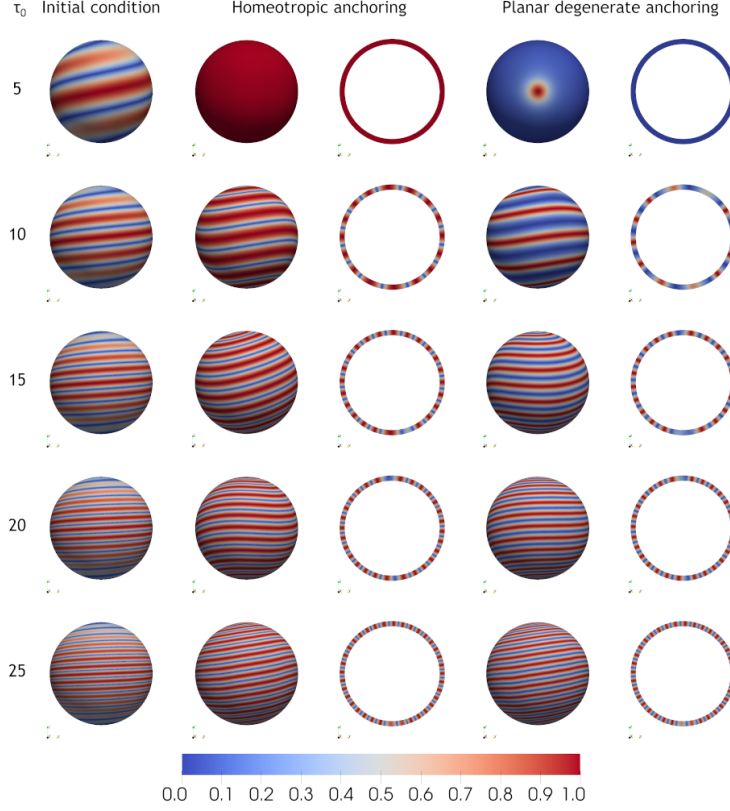


Figure 1: Centered shell ($\Omega_{\text{shell},0}$), front view. Color is proportional to $|\mathbf{n} \cdot \mathbf{r}|$ with $\mathbf{r} = \mathbf{x}/|\mathbf{x}|$. Homeotropic anchoring columns show the outer boundary and a vertical slice through the shell. Planar degenerate anchoring columns have the same format.

2.4 Numerical Simulations

I did a variety of simulations of the cholesteric LdG model on a spherical shell for a range of twist parameters. Our software is implemented in Firedrake, which heavily uses the PETSc library. We used Paraview to visualize our simulations.

The numerical solution for Q is visualized by the following method: We set $\mathbf{n}$ to be the eigenvector of Q corresponding to the largest eigenvalue of Q . If Q has a uniaxial form, then this choice of $\mathbf{n}$ is consistent with (1). Then we compute $|\mathbf{n} \cdot \mathbf{r}|$, where $\mathbf{r}$ is a given vector that corresponds to the computational domain, and visualize $|\mathbf{n} \cdot \mathbf{r}|$ as a scalar field. See Figure 1 for the visualization.

2.5 Studies of defects in liquid crystals

At Carnegie Mellon University, in the summer of 2025 I led an REU and mentored three aspiring undergraduate students. Our goal was the study of so-called “defects” in the director fields τ , χ , and λ .

As seen in Figure 2, τ is the red director, corresponding to $\mathbf{n}$ for any uniaxial $Q = s_0(\mathbf{n} \otimes \mathbf{n} - I/3)$, and χ is the grey director, indicating the axis along which the helical structure rotates. And finally, λ , the blue director, having no physical equivalent, is simply defined to be the director perpendicular to τ and λ .

A defect will occur where τ or λ becomes perpendicular to the outward-pointing radial, i.e. around where $|\tau \cdot \hat{\mathbf{r}}| = 0$ or $|\lambda \cdot \hat{\mathbf{r}}| = 0$. This is the case for a defect of τ or λ , respectively. Defects of τ are equivalent to when $Q = 0$, the isotropic phase. On the other hand, where defects of λ occur, the uniaxial Q will be such that $|\mathbf{n} \cdot \hat{\mathbf{r}}| = 1$, i.e. where the uniaxial Q is aligned with the radial direction. We can give each of these defects a number, which is the winding number of the vector field around such defect. Given

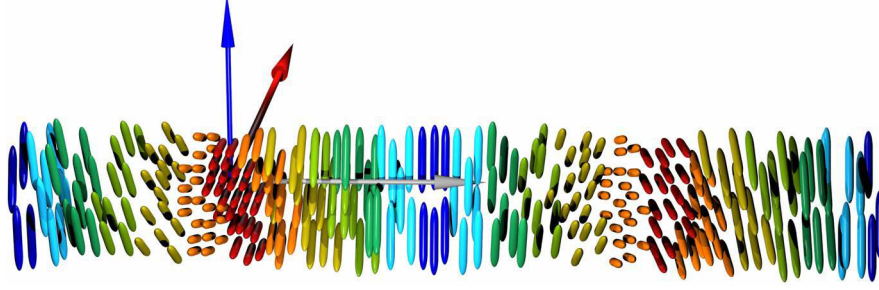


Figure 2: The helical axis structure

certain topological conditions (which can be enforced by sufficiently strong homeotropic anchoring), these defects must add to $+2$.

Figure 3 shows a visualization of positive and negative spirals found in experimentation. Positive spirals are characterized by a $\tau^{+1/2}$ defect at the end of the spiral, and negative spirals by a $\lambda^{+1/2}$ defect at the end of the spiral.

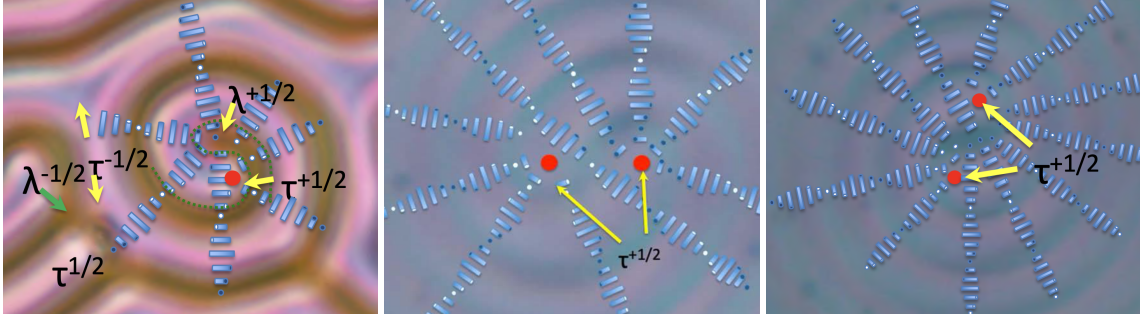


Figure 3: Left: Single clockwise spiral. Middle: Double clockwise spiral. Right: Double counterclockwise spiral.

Several questions about spirals naturally arise based on experimental observation:

1. Why is it that $\tau^{+1/2}$ defects do not separate and spread out on a shell, but come together at the poles and form a spiral pattern? One experiment would be to artificially create four such defects equally spaced on a “great circle,” and see if they come together to form two antipodal spirals. An alternative would be to place the defects in a “tetrahedral” arrangement on the sphere, similar to what happens on a thin nematic shell.
2. The existence of a λ field with defects that add to $+2$ is necessary; where, then, are these defects located on the shell?
3. Why are no $+1$ defects of τ or λ present in either the slab or the shell? Is it because $+1$ defects require a higher energy?

I am currently still working with my students on the first question. The first goal has been to find the appropriate anchoring strength w_0 necessary to give the homeotropic boundary conditions that allow certain topological theorems about the director field to hold. Once these hold, we are simulating the “great circle” and “tetrahedral” initial conditions and seeing what happens with them.

2.6 Future Work

There are several future directions which my research on the LdG model can be taken.

2.6.1 Optimal control problem

The LdG problem may be extended to an optimal control problem. Following [6], this problem is stated as follows. Let time t vary from 0 to a specified final time, t_f . Take $\mathbf{V}$ to be as in the previous section, and introduce the following spaces:

$$\mathcal{C} = \Omega \times (0, t_f), \quad \mathcal{G} = \Gamma \times (0, t_f).$$

We also have target functions:

$$Z_{\mathcal{C}} \in L^2(\mathcal{C}), \quad Z_{\mathcal{G}} \in L^2(\mathcal{G}), \quad Z_{\mathcal{C}} \in H^1(\Omega).$$

The optimal control problem is to minimize the functional:

$$\begin{aligned} J(Q, U_{\mathcal{C}}, U_{\mathcal{G}}) = & \frac{\beta_{\mathcal{C}}}{2} \|Q - Z_{\mathcal{C}}\|_{L^2(\mathcal{C})}^2 + \frac{\beta_{\mathcal{G}}}{2} \|Q - Z_{\mathcal{G}}\|_{L^2(\mathcal{G})}^2 + \frac{\beta_{t_f}}{2} \|Q(\cdot, t_f) - Z_{t_f}(\mathcal{C})\|_{0,\Omega}^2 \\ & + \frac{\alpha_{\mathcal{C}}}{2} \|U_{\Omega}\|_{L^2(\mathcal{C})}^2 + \frac{\alpha_{\mathcal{G}}}{2} \|U_{\Gamma}\|_{L^2(\mathcal{G})}^2, \end{aligned} \quad (9)$$

subject to Q satisfying the strong form PDE.

2.6.2 Experimental validation

Other future research could involve studying other phenomena (besides defects) observed in experimentation. One experiment that might be verified consists of studying an off-centered shell. This shell would be defined by an outer sphere centered at the origin, and an inner sphere centered just above the origin. This would correspond to the effects of gravity pulling down the outer liquid crystal material with respect to the inner aqueous solution. Experiments have shown that, when homeotropic anchoring is applied to the surface, a spiral forms toward the bottom that completely unwinds as one moves to the (thinner) top part of the shell. While some of our numerical experiments have proven promising in terms of reproducing similar results to this, more simulations with constants set and varying the position of the inner sphere, would prove useful in trying to replicate this pattern through an ad-hoc shape optimization.

3 Fluid Flow Through Porous Media

This fall semester at Carnegie Mellon I've started a new research project with Prof. Noel Walkington on fluid flow through porous media. The physics of the problem can be summarized in the following [7].

Consider a porous medium containing N_c conserved components that combine to form N_p phases. Under appropriate assumptions, a weak statement for the balances of mass $\mathbf{m}$ for each component can be written as

$$(\mathbf{m}_t, \hat{\boldsymbol{\mu}})_{\Omega} + a(\nabla \boldsymbol{\mu}, \nabla \hat{\boldsymbol{\mu}}) = 0, \quad (10)$$

for all test functions $\hat{\boldsymbol{\mu}}$, where for $\psi : \mathbb{R}^{N_c} \rightarrow \mathbb{R} \cup \{\infty\}$ convex, proper, and lower semicontinuous, and $a : \mathbb{R}^{N_c \times d} \times \mathbb{R}^{N_c \times d} \rightarrow \mathbb{R}$ is a symmetric positive semi-definite form. We also require that $\boldsymbol{\mu} \in \partial\psi(\mathbf{m})$, i.e. $\boldsymbol{\mu}$ is in the subdifferential of $\psi(\mathbf{m})$, since ψ cannot be guaranteed to be smooth in all places.

An energy balance can be obtained by setting the test function $\hat{\boldsymbol{\mu}} = \boldsymbol{\mu}$ and defining a seminorm via $|\nabla \boldsymbol{\mu}|_a^2 = a(\nabla \boldsymbol{\mu}, \nabla \boldsymbol{\mu})$:

$$\frac{d}{dt} \int_{\Omega} \psi(\mathbf{m}) + |\nabla \boldsymbol{\mu}|_a = 0, \quad (11)$$

Now consider the implicit Euler scheme:

$$(\mathbf{m} - \tilde{\mathbf{m}}, \hat{\boldsymbol{\mu}})_{\Omega} + \delta t a(\nabla \boldsymbol{\mu}, \nabla \hat{\boldsymbol{\mu}}) = 0, \quad (12)$$

with previous iteration $\tilde{\mathbf{m}}$.

This scheme, along with the relation $\boldsymbol{\mu} \in \partial\psi(\mathbf{m})$, can be thought of as a saddle point of the following Lagrangian:

$$L(\boldsymbol{\mu}, \mathbf{m}) = (\mathbf{m} - \tilde{\mathbf{m}}, \boldsymbol{\mu})_{\Omega} + \frac{\delta t}{2} |\nabla \boldsymbol{\mu}|_a^2 - \int_{\Omega} \psi(\mathbf{m}). \quad (13)$$

In particular, we are trying to find the minimum of the function $I(\boldsymbol{\mu})$:

$$I(\boldsymbol{\mu}) = \sup_{\mathbf{m}} L(\boldsymbol{\mu}, \mathbf{m}) = \frac{\delta t}{2} |\nabla \boldsymbol{\mu}|_a^2 - (\tilde{\mathbf{m}}, \boldsymbol{\mu})_{\Omega} + \int_{\Omega} \psi^*(\boldsymbol{\mu}),$$

where $\psi^* : \mathbb{R}^{N_c} \rightarrow \mathbb{R} \cup \{\infty\}$ is the *convex conjugate* of ψ , i.e. the function such that $\int_{\Omega} \psi^*(\boldsymbol{\mu}) + \int_{\Omega} \psi(\mathbf{m}) = (\mathbf{m}, \boldsymbol{\mu})_{\Omega}$.

To minimize $I(\boldsymbol{\mu})$, we will implement a version of Uzawa's algorithm: given an initial guess $\boldsymbol{\mu}_0$, for $k = 0, 1, 2, \dots$ we complete the following two steps:

1. Compute

$$\mathbf{m}_k = \underset{\mathbf{m} \in [0, \infty)^{N_c}}{\operatorname{argmax}} ((\mathbf{m}, \boldsymbol{\mu}_k)_{\Omega} - \int_{\Omega} \psi(\mathbf{m})) \quad (14)$$

2. Find the gradient $\mathbf{g}_k$ of $I(\boldsymbol{\mu})$ at $\boldsymbol{\mu}_k$ by solving

$$\delta t \, a(\nabla \mathbf{g}_k, \nabla \hat{\boldsymbol{\mu}}) + (\mathbf{g}_k, \hat{\boldsymbol{\mu}})_{\Omega} = \delta t \, a(\nabla \boldsymbol{\mu}_k, \nabla \hat{\boldsymbol{\mu}}) + (\mathbf{m}_k - \mathbf{m}_{k-1}, \hat{\boldsymbol{\mu}})_{\Omega}, \quad (15)$$

3. Set $\boldsymbol{\mu}_{k+1} = \boldsymbol{\mu}_k - s_k \mathbf{g}_k$ for some step size $s_k > 0$.

To numerically solve this, a finite difference scheme will be implemented. One important issue with this algorithm is that it is far too slow in practice, and rarely gives meaningful results. My research this year will investigate better ways to implement this algorithm, or alternative approaches.

References

- [1] A. Hicks, Modeling and numerical analysis of the cholesteric landau–de gennes model, Ph.D. thesis. URL https://repository.lsu.edu/gradschool_dissertations/6390
- [2] A. L. Hicks, S. W. Walker, Modelling and simulation of the cholesteric landau-de gennes model, Proceedings of the Royal Society A 480: 20230813 (2024). doi:10.1098/rspa.2023.0813.
- [3] E. G. Virga, Variational Theories for Liquid Crystals, 1st Edition, Vol. 8, Chapman and Hall, London, 1994.
- [4] H. Mori, J. Eugene C. Gartland, J. R. Kelly, P. J. Bos, Multidimensional director modeling using the Q-tensor representation in a liquid crystal cell and its application to the π -cell with patterned electrodes, Japanese Journal of Applied Physics 38 (1R) (1999) 135–146. URL <http://stacks.iop.org/1347-4065/38/i=1R/a=135>
- [5] G. D. Maso, M. Forti, M. Miranda, S. A. Spagnolo, L. Ambrosio (Eds.), Selected Papers, Springer Collected Works in Mathematics, Ennio De Giorgi, 2006.
- [6] T. M. Surowiec, S. W. Walker, Optimal control of the landau–de gennes model of nematic liquid crystals, submitted (2022).
- [7] N. J. Walkington, Notes on solving the porous phase flow problem using uzawa's algorithm.